

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	1	2	3	1	2	1	1	1	1	2	2	1	4	2	2	2	4	2	3	4	3	1	2	3	4	4	3	1	2	2
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	1	4	1	2	3	2	2	2	2	2	2	2	2	1	1	1	1	3	3	1	1	1	2	3	2	4	4	4	4	2
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	1	1	3	4	4	4	1	4	4	4	1	2	1	2	3	3	2	1	2	3	2	1	1	4	3	4	3	3	2	1
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	1	3	4	3	1	3	3	4	2	4	3	1	2	3	2	3	2	4	2	3	3	4	1	2	3	3	4	3	4	3
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	4	3	2	1	1	3	2	2	2	4	1	3	2	1	4	1	3	3	3	3	1	3	4	3	4	2	3	3	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	2	2	4	2	2	4	4	4	3	4	4	2	2	1	2	4	3	1	1	2	2	3	1	2	3	3	1	2	3	3
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	2	3	4	3	3	1	3	1	4	1	2	3	4	1	4	1	4	1	4	1										

HINT - SHEET

SUBJECT : PHYSICS

SECTION - A

1. **Ans (1)**

$$\alpha = {}^4_2\text{He}^{2+} = \text{Helium Nuclei}$$

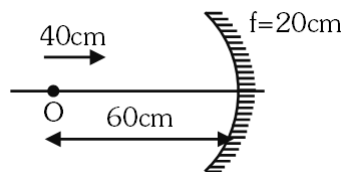
2 protons and 2 neutrons

2. **Ans (2)**

$$E_T = E_0 + E_K$$

$$E_K = E_T - E_0 = 3.555 - 0.51 = 3.045 \text{ MeV}$$

3. **Ans (3)**



$$\frac{dv}{dt} = -\frac{f^2}{(u-f)^2} \frac{du}{dt} = -\frac{(-20)^2}{(-60+20)^2} (40) = -10 \text{ cm/s}$$

speed of image = 10 cm/s

5. **Ans (2)**

Angular width of central maxima

$$\theta_{\text{central}} = \frac{2\lambda}{d}$$

6. **Ans (1)**

$$p = \frac{h}{\lambda} \Rightarrow p \propto \frac{1}{\lambda} \text{ (Rectangular hyperbola)}$$

7. **Ans (1)**

Kinetic energy of thermal neutron with

$$\text{equilibrium is } \frac{3}{2} kT$$

$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2m K.E}} = \frac{h}{\sqrt{2m \left(\frac{3}{2} kT \right)}} = \frac{h}{\sqrt{3 m kT}}$$

8. **Ans (1)**

In semiconductors and insulators, the number of charge carries per unit volume increases with an increase in temperature, so α is negative for these.

9. **Ans (1)**

Here, current flows when any of the switch is in ON (1) state.

A	B	LED
0	0	0
0	1	1
1	0	1
1	1	1

OR Gate

10. **Ans (2)**

As $T \uparrow$, collision $\uparrow$ so relaxation time $\downarrow$ so $v_d \downarrow$

11. **Ans (2)**

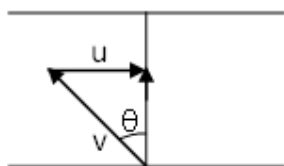
$$\text{Average velocity} = \frac{\text{Total distance}}{\text{Total time}} = \frac{x + 2x}{\frac{x}{v_1} + \frac{2x}{v_2}}$$

$$\Rightarrow v = \frac{3}{\frac{1}{v_1} + \frac{2}{v_2}} \Rightarrow \frac{3}{v} = \frac{1}{v_1} + \frac{2}{v_2}$$

12. **Ans (1)**

$$v = 10 \text{ m/s}$$

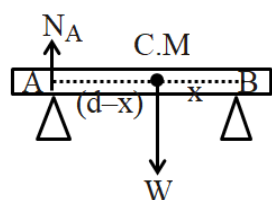
$$u = 5 \text{ m/s}$$



$$\sin \theta = \frac{u}{v} = \frac{10}{20} = \frac{1}{2}$$

$$\Rightarrow \theta = 30^\circ \text{ west}$$

13. **Ans (4)**



$$N_A d = Wx \Rightarrow N_A = \frac{Wx}{d}$$

14. **Ans (2)**

$$a = \frac{g \sin \theta}{1 + K^2/R^2}$$

$$\frac{K^2}{R^2} \text{ is less for hollow sphere.}$$

So, it reaches first.

15. **Ans (2)**

$$\rho_{\text{oil}} h_{\text{oil}} = \rho_{\text{water}} h_{\text{water}}$$

$$\Rightarrow \rho_{\text{oil}} = \frac{1000(15)}{20} = 750 \text{ kg m}^{-3}$$

16. **Ans (2)**

$$\text{At depth : } g_{\text{eff}} = g \left(1 - \frac{d}{R} \right)$$

$$\Rightarrow \frac{g}{n} = g \left(1 - \frac{d}{R} \right) \Rightarrow d = (n - 1) R/n$$

17. **Ans (4)**

$$v_e = \sqrt{\frac{2GM}{R}} = \sqrt{\frac{2G}{R} \times \frac{4}{3} \pi R^3 \rho}$$

$$\text{or } V_e = \sqrt{\frac{8\pi G \rho}{3} R^2}$$

$$\Rightarrow v_e \propto R \Rightarrow \frac{v_e}{v} = \frac{4R}{R} \Rightarrow v_e = 4v$$

18. **Ans (2)**

$$2mv\sqrt{2} = mv_1 \Rightarrow v_1 = 2v\sqrt{2}$$

$$\text{K.E} = \left[\frac{1}{2} (2m) v^2 \right] \times 2 + \frac{1}{2} m (2v\sqrt{2})^2$$

$$= 2mv^2 + 4mv^2 = 6mv^2$$

19. **Ans (3)**

$$a = \omega^2 r = \left(\frac{2\pi}{T} \right)^2 r = \left(\frac{2\pi}{\pi} \right)^2 10$$

$$= 40 \text{ cm/s}^2$$

20. Ans (4)

To complete a vertical circle, speed at A should

$$\text{be } v_A = \sqrt{5gR}$$

$$\text{using energy conservation } mgh = \frac{1}{2}mv_A^2$$

$$h = \frac{1}{2} \frac{v_A^2}{g} = \frac{1}{2} \frac{5g}{g} \frac{D}{2} = \frac{5D}{4}$$

$$\left(R = \frac{D}{2} \right)$$

21. Ans (3)

$$\cos \phi = \frac{R}{Z}$$

$$= \frac{3}{5} = 0.60$$

22. Ans (1)

At resonance

$$w = \frac{1}{\sqrt{LC}}$$

$$w^2 = \frac{1}{LC}$$

$$L = \frac{1}{w^2 C} = \frac{1}{10^6 \times 100 \times 10^{-6}} = 10 \text{ mH}$$

23. Ans (2)

$$\tau = MB \sin 30^\circ \quad \dots(1)$$

$$W = MB (1 - \cos 60^\circ) \quad \dots(2)$$

From (1) and (2)

$$\frac{\tau}{W} = \frac{1/2}{1/2} \Rightarrow \tau = W$$

24. Ans (3)

$$f = \frac{eB}{2\pi m}$$

$$f = \frac{1.76 \times 10^{11} \times 3.57 \times 10^{-2}}{2 \times 3.14} \text{ Hz}$$

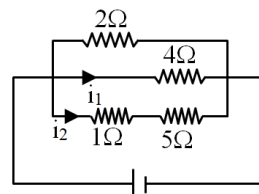
$$f = 10^9 \text{ Hz or } 1 \text{ GHz}$$

25. Ans (4)

$$R' = \frac{\rho \ell'}{A'} = \rho \frac{1.2\ell}{A/1.2} = 1.44 R$$

$$\rho \rightarrow \text{same}$$

26. Ans (4)



$$\frac{i_2}{i_1} = \frac{R_1}{R_2}$$

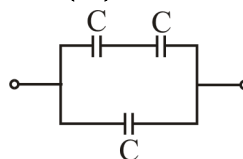
$$\frac{i_2}{3} = \frac{4}{1+5}$$

$$i_2 = 2A$$

$$\text{Power} = P = i_2^2 R$$

$$= 2^2(5) = 20 \text{ watt}$$

27. Ans (3)



$$C_{eq} = \frac{3C}{2} = \frac{3}{2}(6) = 9$$

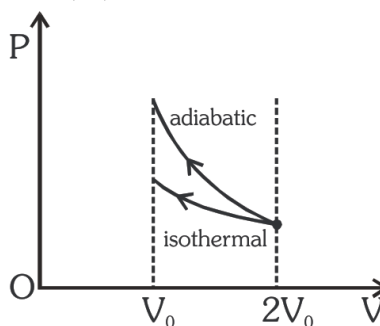
28. Ans (1)

$$v_{rms} \propto \sqrt{T}$$

$$\Rightarrow \frac{v}{200} = \sqrt{\frac{600}{400}}$$

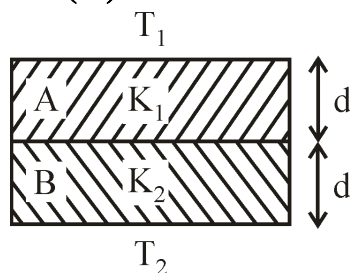
$$\Rightarrow v = 100\sqrt{6}$$

29. Ans (2)



$$W(\text{adiabatic}) > W(\text{isothermal})$$

30. Ans (2)



$$\text{In series, } K_{eq} = \frac{L_1 + L_2}{\frac{L_1}{K_1} + \frac{L_2}{K_2}}$$

$$\frac{d + d}{\frac{d}{K_1} + \frac{d}{K_2}} = \frac{2K_1 K_2}{K_1 + K_2}$$

31. Ans (1)

In adiabatic process

$$Q = 0$$

$$Q = W + \Delta U$$

$$W = -\Delta U$$

when work done by system then work is

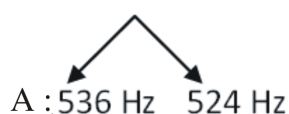
positive so ΔU will be negative

So, temperature will decrease

Here work done by the system will be at
expanse of internal energy

32. Ans (4)

B : 530 Hz



$$\text{Frequency} \propto \sqrt{\text{Tension}}$$

If tension in B is slightly decreased, then

frequency of B decreases and beats increases

therefore frequency of A will be 536 Hz.

33. Ans (1)

$$v \propto \sqrt{\text{Tension}}$$

$$\frac{v_i}{v_f} = \sqrt{\frac{T_i}{T_f}}$$

$$\frac{v_i}{v_f} = \sqrt{\frac{T}{(T/2)}} = \frac{\sqrt{2}}{1}$$

34. Ans (2)

$$f \propto \frac{1}{l}$$

35. Ans (3)

$$LC = 0.01 \text{ cm,}$$

$$ZE = +0.3 \text{ mm} = +0.03 \text{ cm}$$

Measurement :

$$\text{Observed reading} = \text{MSR} + \text{VSR}$$

$$= 9.5 \text{ cm} + (6 \times 0.01) \text{ cm}$$

$$= 9.56 \text{ cm}$$

$$\text{Correct Reading} = \text{Observed reading} - \text{Zero error}$$

$$= 9.56 \text{ cm} - 0.03 \text{ cm} = 9.53 \text{ cm}$$

SECTION - B

36. Ans (2)

Energy released in 1kg of deuteron

$$= \frac{6.02 \times 10^{23} \times 1 \times 10^3}{2} \times 1.6 \text{ MeV}$$

$$= \frac{9.6}{2} \times 10^{32} \text{ eV} \approx 5 \times 10^{32} \text{ eV}$$

37. Ans (2)

$$\frac{1}{F_1} = \frac{1}{f} + \frac{1}{f} \Rightarrow F_1 = f/2 \text{ and } \frac{1}{F_2} = \frac{1}{f} - \frac{1}{f} + \frac{1}{f}$$

$$\Rightarrow F_2 = f \Rightarrow \frac{F_1}{F_2} = \frac{1}{2}$$

38. Ans (2)

For max. intensity

$$I_0 = A_1^2 + A_2^2 + 2A_1 A_2 \cos \phi$$

$$I_0 = (2^2 + 3^2 + 2 \times 2 \times 3) A^2 = 25 A^2$$

$$\text{when path difference is } \frac{\lambda}{2} \{ \cos \phi = \cos 180^\circ = -1 \}$$

$$I = [2^2 + 3^2 + 2(2)(3)(-1)] A^2 = A^2$$

$$\therefore I = \frac{I_0}{25}$$

39. Ans (2)

$$\text{Recoil energy} = \frac{p^2}{2M}$$

$$p = \frac{h}{\lambda} = \frac{hv}{c} \Rightarrow \text{Recoil energy} = \frac{h^2 v^2}{2Mc^2}$$

40. Ans (2)

$$T = 2\pi \sqrt{\frac{\ell}{g}}$$

$$g = \frac{4\pi^2 \ell}{T^2}$$

$$\frac{\Delta g}{g} = \frac{\Delta \ell}{\ell} + \frac{2\Delta T}{T}$$

$$\frac{\Delta g}{g} = \frac{1 \times 10^{-3}}{50 \times 10^{-2}} + 2 \times \frac{1/100}{50/100}$$

$$\frac{\Delta g}{g} = \frac{1}{500} + \frac{2}{50}$$

$$\frac{\Delta g}{g} \times 100 = 0.2 + 4\% = 4.2\%$$

41. Ans (2)

$$100 - T = 0 \Rightarrow T = 100$$

$$T - 0.15 [m + 20] g = 0$$

$$100 - 1.5 [m + 20] = 0$$

$$\frac{15}{10} (m + 20) = 100$$

$$m + 20 = \frac{1000}{15}$$

$$m = \frac{1000 - 300}{15}$$

$$m = \frac{700}{15} = \frac{140}{3} \text{ Kg}$$

42. Ans (2)

$$\omega = \sqrt{\frac{g}{\mu_s r}} = \sqrt{\frac{10}{0.3 \times 2}} = \sqrt{\frac{50}{3}}$$

$$= 4.1 \text{ rad/s}$$

43. Ans (2)

$$(i) W = \Delta K$$

$$\Rightarrow 20 \times 4 + 10 \times 2 = \frac{1}{2} \cdot \frac{1}{2} \cdot V^2$$

$$\Rightarrow V = 20 \text{ m/s}$$

$$(ii) 20 \times 4 + 10 \times 2 + 10 \times 2 - 2 \times 25$$

$$- \frac{1}{2} \times 2 \times 5 = \frac{1}{2} \cdot \frac{1}{2} \cdot V^2$$

$$\Rightarrow 80 + 20 + 20 - 50 - 5 = \frac{V^2}{4}$$

$$\Rightarrow V^2 = 65 \times 4 = 260$$

$$\Rightarrow V = \sqrt{260} = 16.1 \text{ m/s}$$

44. Ans (1)

$$L = \frac{NBA}{l}$$

$$\therefore L = \frac{\mu_0 \mu_r N^2 A}{2\pi R}$$

$$= \frac{4\pi \times 10^{-7} \times 350 \times 500 \times 500 \times 20 \times 10^{-4}}{2\pi \times 40 \times 10^{-2}}$$

$$= 87.5 \text{ mH}$$

45. Ans (1)

$$B = \frac{E}{C}$$

$$\mu_0 H = \frac{E}{C}$$

$$\therefore H = \frac{E}{\mu_0 C} = \frac{(10/\sqrt{2})}{4\pi \times 10^{-7} \times 3 \times 10^8}$$

$$= \frac{1}{12\sqrt{2}\pi} = 18.8 \text{ mili Amp}$$

46. Ans (1)

$$\text{Work} = MB[\cos \theta_1 - \cos \theta_2]$$

$$\text{Work} = MB[\cos 0 - \cos 90^\circ]$$

$$W = NiAB[1 - (0)] = NiAB$$

$$W = 4 \mu\text{J}$$

47. Ans (1)

$$V = \frac{kQ}{r} \text{ \& } E = \frac{kQ}{r^2}$$

$$\Rightarrow V = \sqrt{kQE}$$

48. Ans (3)

$$T = 2\pi\sqrt{\frac{m}{k}}$$

$$4 = 2\pi\sqrt{\frac{m}{k}} \quad \dots\dots(i)$$

$$\text{and } 8 = 2\pi\sqrt{\frac{m+6}{k}} \quad \dots\dots(ii)$$

$$\text{On dividing (i) by (ii), } \left[\frac{1}{2} = \frac{\sqrt{m}}{\sqrt{m+6}} \right]$$

On solving, $m = 2 \text{ kg}$

49. Ans (3)

$$\beta = \frac{Q_2}{W} = \frac{T_L}{T_H - T_L}$$

$$\Rightarrow \frac{600 \times 4.2}{W} = \frac{280}{308 - 280}$$

$$\Rightarrow W = 252 \text{ J}$$

$$\text{Power} = \frac{W}{t} = \frac{252}{1} = 252 \text{ W}$$

50. Ans (1)

Max supply voltage = 10 V

$$V_z = 8 \text{ volt, } V_s = 10 - 8 = 2 \text{ V}$$

$$I_z = \frac{1.6 \text{ W}}{8 \text{ V}} = 0.2 \text{ A,}$$

$$I_L = \frac{8}{10} = 0.8 \text{ A}$$

$$I_s = I_z + I_L = 1 \text{ A}$$

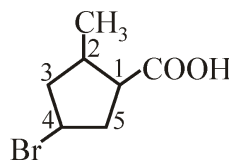
$$\Rightarrow R_s = \frac{V_s}{I_s} = \frac{2}{1} = 2 \text{ } \Omega$$

SUBJECT : CHEMISTRY

SECTION - A

57. Ans (4)

NCERT-XI, Pg. # 268, Part-II, Ed.23-24



4-Bromo-2-methylcyclopentane carboxylic acid

60. Ans (2)

NCERT-XI, Pg. # 261, Part-II, Ed 23-24

62. Ans (1)

NCERT-XI, Pg. # 308, Part-II, Ed. 23-24

74. Ans (2)

$$\Delta T_f = k_f \times m \times i \quad (\text{Assume } m \simeq M)$$

$$\downarrow \Delta T_f \propto (M \times i) \downarrow$$

$$0.1 \text{ M NaCl} \Rightarrow 0.1 \times 2 = 0.2$$

$$0.05 \text{ M K}_2\text{SO}_4 \Rightarrow 0.05 \times 3 = 0.15 \text{ (minimum)}$$

$$0.1 \text{ M CaCl}_2 \Rightarrow 0.1 \times 3 = 0.3$$

$$0.2 \text{ M KCl} \Rightarrow 0.2 \times 2 = 0.4$$

75. Ans (3)

$$\pi = CRT$$

$$= \frac{10}{200000} \times \frac{1000}{500} \times 0.0821 \times 300 \text{ atm}$$

$$= \frac{0.2463}{100} \text{ atm}$$

$$= \frac{0.2463}{100} \times 1.01325 \times 10^5 \text{ Pa}$$

$$= 249.56 \text{ Pa}$$

76. Ans (3)

For zero order reaction, $r = k$

$$r = \frac{1}{3} \frac{d[\text{H}_2]}{dt} = k$$

$$k = \frac{7.5 \times 10^{-4}}{3} \text{ Ms}^{-1}$$

$$= 2.5 \times 10^{-4} \text{ Ms}^{-1}$$

77. **Ans (2)**

NCERT-XII, Pg. # 72, 86, 87, 88, Part-1, Edition-2017

78. **Ans (1)**

$$\frac{w}{E} = \frac{Q}{F}$$

$$\frac{w}{108/1} = \frac{2 \times 9.65 \times 60}{96500}$$

$$w = 1.296 \text{ g}$$

79. **Ans (2)**

NCERT-XII, Pg. # 72, Part-1, Edition 2017

80. **Ans (3)**

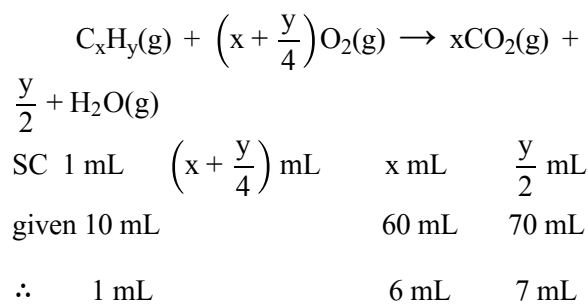
NCERT-XI, Pg. # 219, Part-1

For basic buffer solution

$$\text{pOH} = \text{pK}_b + \log \frac{[\text{C}_5\text{H}_5\text{N}^+\text{Cl}^-]}{[\text{C}_5\text{H}_5\text{N}]}$$

$$\text{pH} = 14 - \text{pOH}$$

81. **Ans (2)**



On comparing, we get

$$x = 6, \frac{y}{2} = 7 \Rightarrow y = 14$$

Formula of hydrocarbon $\Rightarrow \text{C}_6\text{H}_{14}$

$$\therefore x + y = 6 + 14 = 20$$

82. **Ans (1)**

NCERT-XI, Pg. # 244, Part-1

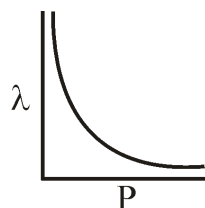
Disproportionation reaction is that redox reaction in which same element is getting oxidized as well as reduced simultaneously.

83. **Ans (1)**

NCERT-XI, Pg. # 46, Part-1

$$\therefore \lambda = \frac{h}{P} \left(\lambda \propto \frac{1}{P} \right)$$

$$\Rightarrow \lambda P = h \text{ (constant)}$$



So plot will be a rectangular hyperbola.

84. **Ans (4)**

NCERT-XI, Pg. # 44, Part-1

85. **Ans (3)**

NCERT-XI, Pg. # 170, 171, 172, 173, Part-1

SECTION - B

88. **Ans (3)**

NCERT-XI, Pg. # 273, Part-(ii) Ed, 23-24

89. **Ans (2)**

NCERT-XII, Pg. # 269, Part-II, Ed. 23-24

90. **Ans (1)**

NCERT-XI, Pg. # 274, Part-II, Ed. 23-24

96. **Ans (3)**

NCERT-XII, Pg. # 110, 116, Part-I, Edition 2017

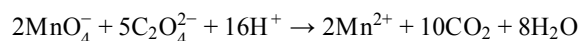
If one reactant is present in excess among two reactants in a chemical reaction then order can altered.

Decomposition of gaseous NH_3 on hot Pt surface.

(i) at high pressure $\Rightarrow$ Zero order reaction

(ii) at low pressure $\Rightarrow$ first order reaction

97. **Ans (3)**



(pink colour is due to excess of unreacted KMnO_4)

98. **Ans (4)**

NCERT-XI, Pg. # 204, 201, 205, 202, Part-1

99. **Ans (2)**

NCERT-XI, Pg. # 179, Part-1

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ = -2.303 RT \log K$$

$$\Delta G^\circ = 50 \times 10^3 - 300 \times 20$$

$$= -2.303 \times 8.314 \times 300 \log K$$

$$\log K = -7.65$$

On taking antilog

$$K = 2.2 \times 10^{-8}$$

100. **Ans (4)**

NCERT-XI, Pg. # 247, Part-1

SUBJECT : BOTANY

SECTION - A

101. **Ans (3)**

NCERT XII Pg.# 33(E), 34(H)

102. **Ans (1)**

NCERT XII Pg.# 34(E), 36(H)

103. **Ans (2)**

NCERT XI Pg.# 74

104. **Ans (3)**

NCERT XI Pg.# 68

105. **Ans (2)**

NCERT XI Pg.# 93,94

106. **Ans (3)**

NCERT XI Pg.# 89

107. **Ans (2)**

NCERT (XII) Pg # 106

108. **Ans (4)**

NCERT (XII) Pg # 114

109. **Ans (2)**

NCERT (XII) Pg # 112

110. **Ans (3)**

NCERT (XII) Pg # 112

111. **Ans (3)**

NCERT (XII) Pg # 86

112. **Ans (4)**

NCERT (XII) Pg # 89

113. **Ans (1)**

NCERT (XII) Pg # 83

114. **Ans (2)**

NCERT XI Pg. # 250

115. **Ans (3)**

NCERT XI Pg. # 228, 229

116. **Ans (3)**

NCERT XI Pg. # 219 (Fig. 13.9)

117. **Ans (4)**

NCERT XI Pg. # 248

118. **Ans (3)**

NCERT XI Pg. # 222

119. **Ans (4)**

NCERT XI Pg. # 158, 159

120. **Ans (3)**

NCERT XI Pg. # 230

121. **Ans (2)**

NCERT-XI, Pg.# 38

122. **Ans (4)**

NCERT-XI, Pg.# 262

123. **Ans (3)**
NCERT-XII, Pg.# 266
124. **Ans (2)**
NCERT-XII Pg.# 233
125. **Ans (1)**
NCERT-XII, Pg.# 228
126. **Ans (1)**
NCERT-XI Pg.# 32
127. **Ans (3)**
NCERT-XII, Pg.# 246
128. **Ans (2)**
NCERT-XII, Pg.# 265, 266, 267
129. **Ans (2)**
NCERT XIth Pg. # 23
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